

EXPERIMENT - 3

AIM

To create an Employee–Department–Project relational database, insert at least 30 employees across 5 departments and 8 projects, and demonstrate SQL queries using Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE expressions, and ORDER BY.

SOURCE CODE

```
CREATE DATABASE CompanyDB;
```

```
USE CompanyDB;
```

```
CREATE TABLE Department (
```

```
    dept_id INT PRIMARY KEY,
```

```
    dept_name VARCHAR(50) NOT NULL
```

```
);
```

```
CREATE TABLE Project (
```

```
    project_id INT PRIMARY KEY,
```

```
    project_name VARCHAR(100) NOT NULL,
```

```
    budget DECIMAL(12,2),
```

```
    dept_id INT,
```

```
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
```

```
);
```

```
CREATE TABLE Employee (
```

```
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(50) NOT NULL,
```

```
    salary DECIMAL(10,2),
```

```
hire_date DATE,

dept_id INT,

project_id INT,

FOREIGN KEY (dept_id) REFERENCES Department(dept_id),

FOREIGN KEY (project_id) REFERENCES Project(project_id)
);
-- Insert Departments

INSERT INTO Department VALUES

(1,'IT'),(2,'HR'),(3,'Finance'),(4,'Marketing'),(5,'Operations');

-- Insert Projects

INSERT INTO Project VALUES

(101,'Cloud Migration',150000,1),

(102,'AI Analytics',200000,1),

(103,'Recruitment Portal',80000,2),

(104,'Financial Dashboard',120000,3),

(105,'Digital Campaign',90000,4),

(106,'Supply Chain System',180000,5),

(107,'Mobile Application',140000,1),

(108,'Employee Wellness',60000,2);

-- Insert 30 Employees

INSERT INTO Employee VALUES

(1,'Aarav',65000,'2022-01-10',1,101),

(2,'Vivaan',72000,'2021-03-15',1,102),
```

(3,'Aditya',58000,'2023-06-20',1,107),
(4,'Arjun',81000,'2020-08-12',1,101),
(5,'Kabir',69000,'2022-11-05',1,102),
(6,'Reyansh',62000,'2024-02-18',1,107),
(7,'Ananya',52000,'2022-04-11',2,103),
(8,'Diya',48000,'2023-01-22',2,108),
(9,'Myra',61000,'2021-07-19',2,103),
(10,'Sara',55000,'2022-09-30',2,108),
(11,'Ishita',47000,'2024-03-14',2,103),
(12,'Meera',59000,'2020-12-01',2,108),
(13,'Rohan',75000,'2021-02-17',3,104),
(14,'Karan',68000,'2022-05-09',3,104),
(15,'Nikhil',83000,'2019-10-21',3,104),
(16,'Yash',62000,'2023-08-13',3,104),
(17,'Manav',71000,'2021-11-28',3,104),
(18,'Dev',56000,'2024-01-09',3,104),
(19,'Aanya',54000,'2022-02-25',4,105),
(20,'Kiara',63000,'2021-06-16',4,105),
(21,'Tanya',57000,'2023-04-10',4,105),
(22,'Riya',66000,'2020-09-07',4,105),
(23,'Avni',51000,'2024-02-05',4,105),
(24,'Navya',70000,'2022-12-19',4,105),

```
(25,'Samar',60000,'2021-01-12',5,106),
(26,'Dhruv',73000,'2020-04-23',5,106),
(27,'Atharv',67000,'2022-07-18',5,106),
(28,'Parth',59000,'2023-05-29',5,106),
(29,'Rudra',76000,'2019-08-31',5,106),
(30,'Veer',64000,'2024-01-20',5,106);
```

SQL QUERIES

1. Selection

Display employees whose salary is greater than 70,000.

```
SELECT * FROM Employee WHERE salary > 70000;
```



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I/O

AI Agent

102 ms

STDIN

Input for the program (Optional)

Output:

emp_id	emp_name	salary	hire_date	dept_id	project_id
2	Vivaan	72000.00	2021-03-15	1	102
4	Arjun	81000.00	2020-08-12	1	101
13	Rohan	75000.00	2021-02-17	3	104
15	Nikhil	83000.00	2019-10-21	3	104
17	Manav	71000.00	2021-11-28	3	104
26	Dhruv	73000.00	2020-04-23	5	106
29	Rudra	76000.00	2019-08-31	5	106

2. Projection

Display only employee names and salaries.

```
SELECT emp_name, salary FROM Employee;
```

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AI Agent

125 ms

STDIN

Input for the program (Optional)

Output:

emp_name	salary
Aarav	65000.00
Vivaan	72000.00
Aditya	58000.00
Arjun	81000.00
Kabir	69000.00
Reyansh	62000.00
Ananya	52000.00
Diya	48000.00
Myra	61000.00
Sara	55000.00
Ishita	47000.00
Meera	59000.00
Rohan	75000.00
Karan	68000.00
Nikhil	83000.00
Yash	62000.00
Manav	71000.00
Dev	56000.00
Aanya	54000.00
Kiara	63000.00
Tanya	57000.00
Riya	66000.00
Avni	51000.00
Navya	70000.00
Samar	60000.00
Dhruv	73000.00
Atharv	67000.00
Parth	59000.00
Rudra	76000.00
Veer	64000.00

Light

3. Aggregate Functions

Find total employees, average salary, maximum salary, minimum salary and total salary.

```
SELECT COUNT(*) AS total_employees,
```

```
    AVG(salary) AS average_salary,
```

```
    MAX(salary) AS highest_salary,
```

```
    MIN(salary) AS lowest_salary,
```

```
    SUM(salary) AS total_salary
```

```
FROM Employee;
```



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AI Agent

119 ms

STDIN

Input for the program (Optional)

Output:

total_employees	average_salary	highest_salary	lowest_salary	total_salary
30	63466.666667	83000.00	47000.00	1904000.00

4. GROUP BY

Display the number of employees and average salary in each department.

```
SELECT d.dept_name,
```

```
    COUNT(e.emp_id) AS employee_count,
```

```
    AVG(e.salary) AS average_salary
```

```
FROM Department d
```

JOIN Employee e ON d.dept_id = e.dept_id

GROUP BY d.dept_id, d.dept_name;

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 I/O  AI Agent 85 ms 

STDIN

Input for the program (Optional)

Output:

dept_name	employee_count	average_salary
IT	6	67833.333333
HR	6	53666.666667
Finance	6	69166.666667
Marketing	6	60166.666667
Operations	6	66500.000000

5. HAVING

Display departments whose average salary is greater than 65,000.

```
SELECT d.dept_name,  
AVG(e.salary) AS average_salary  
FROM Department d  
JOIN Employee e ON d.dept_id = e.dept_id  
GROUP BY d.dept_id, d.dept_name  
HAVING AVG(e.salary) > 65000;
```



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I/O

AI Agent

88 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+-----+
| dept_name | average_salary |
+-----+-----+
| IT        | 67833.333333   |
| Finance   | 69166.666667   |
| Operations | 66500.000000   |
+-----+-----+
```

6. CASE Expression

Classify employees according to their salary.

```
SELECT emp_name, salary,
CASE
  WHEN salary >= 75000 THEN 'High Salary'
  WHEN salary >= 60000 THEN 'Medium Salary'
  ELSE 'Low Salary'
END AS salary_category
FROM Employee;
```



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I/O

AI Agent

134 ms

STDIN

Input for the program (Optional)

Output:

emp_name	salary	salary_category
Aarav	65000.00	Medium Salary
Vivaan	72000.00	Medium Salary
Aditya	58000.00	Low Salary
Arjun	81000.00	High Salary
Kabir	69000.00	Medium Salary
Reyansh	62000.00	Medium Salary
Ananya	52000.00	Low Salary
Diya	48000.00	Low Salary
Myra	61000.00	Medium Salary
Sara	55000.00	Low Salary
Ishita	47000.00	Low Salary
Meera	59000.00	Low Salary
Rohan	75000.00	High Salary
Karan	68000.00	Medium Salary
Nikhil	83000.00	High Salary
Yash	62000.00	Medium Salary
Manav	71000.00	Medium Salary
Dev	56000.00	Low Salary
Aanya	54000.00	Low Salary
Kiara	63000.00	Medium Salary
Tanya	57000.00	Low Salary
Riya	66000.00	Medium Salary
Avni	51000.00	Low Salary
Navya	70000.00	Medium Salary
Samar	60000.00	Medium Salary
Dhruv	73000.00	Medium Salary
Atharv	67000.00	Medium Salary
Parth	59000.00	Low Salary
Rudra	76000.00	High Salary
Veer	64000.00	Medium Salary

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7. ORDER BY

Display employees in descending order of salary.

```
SELECT emp_name, salary
```

```
FROM Employee
```

```
ORDER BY salary DESC;
```



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I/O

AI Agent

109 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+-----+
| emp_name | salary |
+-----+-----+
| Nikhil   | 83000.00 |
| Arjun    | 81000.00 |
| Rudra    | 76000.00 |
| Rohan    | 75000.00 |
| Dhruv    | 73000.00 |
| Vivaan   | 72000.00 |
| Manav    | 71000.00 |
| Navya    | 70000.00 |
| Kabir    | 69000.00 |
| Karan    | 68000.00 |
| Atharv   | 67000.00 |
| Riya     | 66000.00 |
| Aarav    | 65000.00 |
| Veer     | 64000.00 |
| Kiara    | 63000.00 |
| Yash     | 62000.00 |
| Reyansh  | 62000.00 |
| Myra     | 61000.00 |
| Samar    | 60000.00 |
| Meera    | 59000.00 |
| Parth    | 59000.00 |
| Aditya   | 58000.00 |
| Tanya    | 57000.00 |
| Dev      | 56000.00 |
| Sara     | 55000.00 |
| Anya     | 54000.00 |
| Ananya   | 52000.00 |
| Avni     | 51000.00 |
| Diya     | 48000.00 |
| Ishita   | 47000.00 |
+-----+-----+
```

8. JOIN Employee, Department and Project

Display employee name, department, project and salary.

```
SELECT e.emp_name, d.dept_name, p.project_name, e.salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
JOIN Project p ON e.project_id = p.project_id
ORDER BY d.dept_name, e.emp_name;
```



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AI Agent

91 ms

STDIN

Input for the program (Optional)

Output:

emp_name	dept_name	project_name	salary
Dev	Finance	Financial Dashboard	56000.00
Karan	Finance	Financial Dashboard	68000.00
Manav	Finance	Financial Dashboard	71000.00
Nikhil	Finance	Financial Dashboard	83000.00
Rohan	Finance	Financial Dashboard	75000.00
Yash	Finance	Financial Dashboard	62000.00
Ananya	HR	Recruitment Portal	52000.00
Diya	HR	Employee Wellness	48000.00
Ishita	HR	Recruitment Portal	47000.00
Meera	HR	Employee Wellness	59000.00
Myra	HR	Recruitment Portal	61000.00
Sara	HR	Employee Wellness	55000.00
Aarav	IT	Cloud Migration	65000.00
Aditya	IT	Mobile Application	58000.00
Arjun	IT	Cloud Migration	81000.00
Kabir	IT	AI Analytics	69000.00
Reyansh	IT	Mobile Application	62000.00
Vivaan	IT	AI Analytics	72000.00
Aanya	Marketing	Digital Campaign	54000.00
Avni	Marketing	Digital Campaign	51000.00
Kiara	Marketing	Digital Campaign	63000.00
Navya	Marketing	Digital Campaign	70000.00
Riya	Marketing	Digital Campaign	66000.00
Tanya	Marketing	Digital Campaign	57000.00
Atharv	Operations	Supply Chain System	67000.00
Dhruv	Operations	Supply Chain System	73000.00
Parth	Operations	Supply Chain System	59000.00
Rudra	Operations	Supply Chain System	76000.00
Samar	Operations	Supply Chain System	60000.00
Veer	Operations	Supply Chain System	64000.00

RESULT

The Employee–Department–Project database was successfully created with:

- 30 Employees
- 5 Departments
- 8 Projects

The following SQL concepts were successfully demonstrated:

- Selection
- Projection
- Aggregate Functions
- GROUP BY
- HAVING
- CASE Expression
- ORDER BY
- JOIN